

PAPER COPILOT: TRACKING THE EVOLUTION OF PEER REVIEW IN AI CONFERENCES

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ABSTRACT

The rapid growth of AI conferences is straining an already fragile peer-review system, leading to heavy reviewer workloads, expertise mismatches, inconsistent evaluation standards, superficial or templated reviews, and limited accountability under compressed timelines. In response, conference organizers have introduced new policies and interventions to preserve review standards. Yet these ad-hoc changes often create further concerns and confusion about the review process, leaving how papers are ultimately accepted—and how practices evolve across years—largely opaque. We present PAPER COPILOT, a system that creates durable digital archives of peer reviews across a wide range of computer-science venues, an open dataset that enables researchers to study peer review at scale, and a large-scale empirical analysis of ICLR reviews spanning multiple years. By releasing both the infrastructure and the dataset, PAPER COPILOT supports reproducible research on the evolution of peer review. We hope these resources help the community track changes, diagnose failure modes, and inform evidence-based improvements toward a more robust, transparent, and reliable peer-review system.

1 INTRODUCTION

The rapid growth of submissions to top-tier Artificial Intelligence (AI) and Machine Learning (ML) conferences—now exceeding 10,000 per venue annually (Yang, 2025)—has placed unprecedented pressure on peer review. To address issues of scale and transparency, many conferences have adopted open or semi-open platforms like OpenReview. Yet practices remain inconsistent: some venues publish reviews and scores, while others keep them private. This inconsistency has sparked ongoing debate around fairness, accountability, and the effectiveness of different review models (Cortes & Lawrence, 2021).

Meanwhile, the nature of review data itself has evolved. Whereas reviews were initially limited to a single numerical rating, nowadays reviews include multiple dimensions such as soundness, technical correctness, and presentation quality (Beygelzimer et al., 2021). Since the rise of OpenReview (Soergel et al., 2013) and especially following the widespread adoption of formal rebuttal phases, the need to analyze score dynamics over time has grown (Yang, 2025). Yet community discourse remains scattered—spread across Twitter, Reddit, Zhihu, Xiaohongshu, and other social platforms in fragmented, venue-specific threads. As submission volumes and review complexity continue to increase, the absence of structured tools that unify and visualize the review data has become a bottleneck for both transparency and timely author decision-making.

From the standpoint of conference organizers, internal dashboards provide visibility into score trends and acceptance distributions (Cortes & Lawrence, 2021). But these views are rarely made accessible to authors. During the critical post-review phase—often only one or two weeks long—authors are expected to analyze reviewer feedback, prepare rebuttals, and potentially decide whether to withdraw or revise their submissions. With scattered resources, authors must resort to scraping statistics (Sun, 2020) or aggregating posts from scattered social media. This inefficiency hinders timely decision-making and may even lead to missed opportunities for meaningful rebuttal or clarification. A system that actively collects review statistics, visualizes score dynamics, and contextualizes submissions relative to their peers would therefore offer immediate utility to thousands of authors.

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In addition, while the OpenReview ecosystem has grown, many conferences still use closed-form review systems and opt not to release reviews or scores publicly. Motivations for withholding reviews may include protection of intellectual content, reviewer anonymity, or historical precedent. However, survey studies of peer review in ML have documented that such opacity exacerbates known problems: inconsistent reviewing standards, lack of calibration across reviewers, and limited accountability for low-quality or biased reviews (Shah, 2022; Sun et al., 2025). These shortcomings directly affect fairness and community trust, and they compound when authors have no visibility into aggregate statistics. A practical workaround is to allow voluntary, anonymized community submissions. If designed carefully, such a system can extract value from partially open data while respecting privacy and consent.

Beyond the review process itself, the AI / ML community lacks robust infrastructure for tracking who is shaping the field over time. Traditional platforms like Google Scholar and DBLP focus primarily on paper-level or author-level metadata—citations, references, and publication profiles (Scholar; DBLP). While Google Scholar does display author affiliations (and, by extension, country information), these systems were not designed to support temporal analysis or dynamic tracking across institutions or regions. Their scope remains limited, offering little visibility into broader affiliation-, institution-, or country-level dynamics. Tracking such information has historically been left to organizations due to the technical and logistical challenges involved, as seen in efforts like CSRankings (Berger) or university rankings.

As AI becomes increasingly high-profile and competitive, attention is increasingly concentrated around specific venue cycles. This shift highlights a growing need to understand who is actively driving progress over shorter timeframes: which institutions are rising, who remains active in the field, and what geographic regions are gaining influence. Yet no existing academic tool provides this kind of dynamic, multi-scale view. Ranking systems often update annually and draw from non-transparent data sources, making them ill-suited to capture the rapid and evolving dynamics of the field.

To address these needs and challenges, we introduce **Paper Copilot**, a system with open dataset for tracking peer review dynamics and talent trajectories across AI/ML. Our contributions include:

- **System for tracking AI/ML progress.** A venue-configurable pipeline (Fig. 1) that unifies multi-source inputs into standardized, versioned paperlists and powers interactive, multi-granularity analytics (venue/institution/country) for longitudinal progress tracking.
- **Scalable peer-review archive.** A unified archive built from open, semi-open, and opt-in community data with temporal snapshots, capturing multi-dimensional review metadata and daily dynamics (ICLR 2024/2025), to preserve and analyze cross-venue review evolution at scale.
- **Findings & ethical considerations.** Empirical results show a 2025 shift toward sharper, score-driven tiering under volume pressure and characteristic rebuttal-phase dynamics (score shifts, consensus evolution); we articulate safeguards on sourcing/consent, privacy, misuse, and bias.

2 RELATED WORKS

2.1 PEER REVIEW DATASETS AND ANALYSIS

There is a growing body of work on creating datasets to study academic peer review. PeerRead (Kang et al., 2018), collected 14.7K paper drafts with reviews and decisions from NLP / ML venues including ACL, NeurIPS and ICLR, enabling research on review–decision alignment and NLP applications like review score prediction. More recently, MOPRD (Lin et al., 2023) introduced a multidisciplinary open peer review dataset spanning several journals and computer science conferences, capturing the full reviewing process (including review comments, rebuttals, meta-reviews and final decisions) across domains. Other efforts have targeted specific aspects: for example, ORB (Szumega et al., 2023) curated a list of more than 36,000 scientific papers with their more than 89,000 reviews and final decisions in high energy physics.

Bharti et al. (2022) leveraged an NLP approach to estimate reviewer confidence from review text, illustrating analysis that can be done when reviews are available. Peer Review Analysis (Ghosal et al., 2022) provided 1,199 review reports labeled at the sentence level for aspects like critique, suggestion,

and sentiment, serving as a benchmark for automated review assessment. Ribeiro et al. (2021) used a private dataset of two non-disclosed conferences to perform RSP, PDP, and sentiment analysis to extract review polarities. In general, these datasets have been invaluable in enabling studies on review quality, bias, and consistency. However, each is often limited to a particular venue or a snapshot in time. We advance beyond prior datasets by unifying multiple venues over a long temporal window, thereby supporting cross-conference and longitudinal analyses (e.g., how review score distributions shift over years, or how an author’s successive submissions fare over time). It also introduces the notion of tracking review process “dynamics,” since our data include timestamps and sequences (allowing one to reconstruct the timeline of reviews and discussions for open-review conferences). This complements earlier one-off studies that examined review consistency and randomness in single years (Beygelzimer et al., 2021; Cortes & Lawrence, 2021), by providing a broader data foundation to study such effects in the aggregate.

2.2 PEER REVIEW TRACKING SYSTEMS

Our work is related to platforms that manage or expose the peer review process. OpenReview.net is the most prominent system enabling open peer review for many ML conferences (ICLR, NeurIPS, UAI, etc.), providing public APIs to fetch submissions, reviews, and comments (OpenReview, 2024). We extensively leverage OpenReview as a data source, but whereas OpenReview is focused on facilitating the review process for active conferences, our system is designed to track and analyze review outcomes across conferences and years. In some sense, Paper Copilot serves as a meta-layer on top of systems like OpenReview and other open-access portals and proceedings (Foundation, 2024; Anthology), aggregating their outputs for comparative analysis. A few individual conferences have released summary statistics (e.g., acceptance rates, score distributions) in blog posts or slides, but these are ad-hoc and not standardized (Beygelzimer et al., 2021). To our knowledge, there was no public web resource before Paper Copilot that allowed users to conveniently browse and compare peer review results (scores, decisions, etc.) for a wide range of AI / ML venues in one place. Our system fills this gap by acting as a centralized peer review statistics dashboard for the community. It is somewhat analogous to “CS Conference Stats” websites that track acceptance rates (Berger), but we go further by including detailed review metrics and providing the underlying dataset.

2.3 AUTHOR PROFILING AND METASCIENCE

Studying author career trajectories and publication patterns has been a long-standing interest in bibliometrics (Fan et al., 2024; De Frutos Belizon et al., 2025). Large-scale academic databases and knowledge graphs – such as Google Scholar, Microsoft Academic Graph, Semantic Scholar’s Open Research Corpus, and AMiner – have enabled analyses of collaboration networks, citation trajectories, and researcher mobility (Scholar). For example, AMiner (Tang et al., 2008), an academic social network mining system that compiled tens of thousands of author profiles and their affiliations over time, allowing for queries on career moves and productivity. A recent dataset from AMiner contains over 57,000 scholars’ career paths (with 42,230 affiliations) for studying academic job transitions. These tools, however, focus on publication and citation data; they generally do not incorporate the peer review stage. We bring a new dimension by linking authors to the review process outcomes of their submissions. This creates opportunities for “talent trajectory” studies in AI / ML – for instance, one could track how an early-career researcher improves their paper quality (as reflected in review scores) over successive submissions, or examine whether consistent high reviewers’ ratings correlate with future citation impact (Vinkenburg et al., 2020). Our dataset, combined with external bibliometric data, could also help identify patterns such as mentorship lineages (e.g., how students of certain groups perform in reviews) or geographic trends in research feedback (Guevara et al., 2016). In the broader context of metascience, our work contributes an open resource to empirically study questions about peer review and researcher development. This aligns with recent calls for transparency and data-driven policy in peer review (Bianchi & Squazzoni, 2022). By making our data easily accessible, we hope to facilitate further meta-research on how review processes influence and reflect the growth of the AI / ML research community.

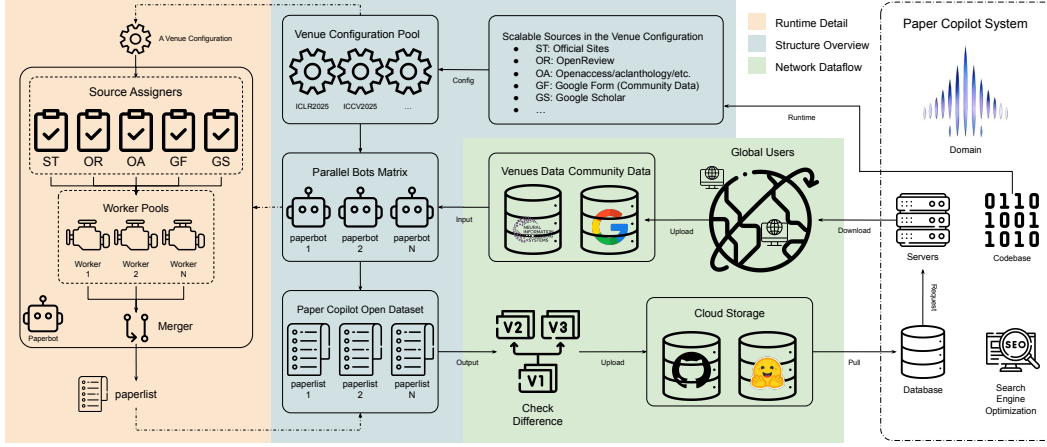


Figure 1: Paper Copilot System Architecture

3 SYSTEM

Paper Copilot is a modular system for large-scale collection and presentation of peer review dynamics and talent trajectories. Figure 1 outlines its architecture: venue configurations define scalable sources, processed through assigners, worker pools, and parallel bots to generate standardized paperlists. Versioned datasets are validated, uploaded to cloud storage, and served globally through optimized backend, database, and frontend components. The following sections detail data collection, system backend/frontend, and maintenance.

Data Collection We use a hybrid data ingestion pipeline to accommodate diverse conference policies, including: 1) *OpenReview API*: For open-review venues (e.g., ICLR, NeurIPS), scheduled scripts pull submission metadata, review scores, confidences, and discussion threads. We store timestamped snapshots to track score changes (e.g., pre-/post-rebuttal). 2) *Website Scraping*: For venues without APIs (e.g., CVPR, AAAI), lightweight scrapers extract accepted papers, authors, and other publicly available metadata from official websites or proceedings. 3) *Community Contributions*: For closed-review venues, authors voluntarily submit reviews via a structured form, yielding 6,584 valid reviews per submission to date.

All data flows through a cleaning and normalization module that standardizes fields, deduplicates entries, and prepares them for integration. Adding new venues requires minimal configuration.

Backend and Frontend The backend is built on a LAMP stack (Linux, Apache, MySQL, PHP) and deployed via a cPanel-managed hosting environment, which simplifies cron scheduling, backups, and database management. Conference data—submissions, reviews, authorship—is stored in a MySQL relational database with a normalized schema linking papers, reviews, and authors. Pre-computed aggregates (e.g., average score, acceptance rate per venue/year) are cached to support fast retrieval for analytics and display.

The user-facing website is built with WordPress, extended by custom PHP and JavaScript for dynamic rendering. All pages support filtering by score range, decision status, or specific metadata fields, and provide links to external resources such as OpenReview, official proceedings, or ArXiv when available. Each conference currently has two primary views: 1) *Statistics*: Interactive charts display trends in paper counts, acceptance rates, score distributions, and reviewer score rankings. 2) *Paper list*: Tabulated entries show paper titles, primary areas, authors, affiliations, departments, countries, citations, and acceptance types (e.g., oral/poster).

Beyond individual venues, the system supports cross-venue, longitudinal analysis at the institutional level. Users can explore hierarchical structures—tracking contributions over time from institutions to departments, laboratories, and regions. This enables global and fine-grained insights into talent flows, geographic shifts, and institutional participation across the field. For more advanced exploration, we provide a companion search tool that can run locally. Users can query by score thresholds, keywords, or metadata fields, operating on either the live database or JSON exports.